

Chocolate BI

Sector: Retail / FMCG

Team ID: G-17 | Section: Section B

Newton School of Technology

Chocolate Sales Performance Analysis & Business Intelligence

Project Lead: [Yashkumar Nimje]

Analysis Lead: [Shivansh Upadhyay]

Strategy Lead: [Manpreet Singh]

Data Lead: [Shane Joseph]

Visualization Lead: [Saksham Narotra]

PPT & Quality Lead: [Ashar Shaikh]

Faculty Mentor: [satyaki sir]



Capstone G-17

GitHub Repository

https://github.com/mansingh-04/SectionB_G-17_ChocolateSalesDataset

Tableau Dashboard

https://public.tableau.com/views/Chocolate_Sales_G17/SalesDashboard?:language=en-US&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

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Context & Problem Statement

Sector Context

- Global chocolate confectionery market is highly competitive
- Sales performance varies across geographies, product lines, store formats, and customer segments
- Retail decision-makers need a clear, data-driven view of revenue sources, underperforming products, and high-value segments

Problem Statement

Which combination of product category, store type, and customer segment drives the highest revenue per transaction—and where are the biggest untapped growth opportunities?

Objective

Strategy 01

Reallocate promotional budgets to highest-ROI product–store–customer combinations

Strategy 02

Discontinue or reposition low-performing SKUs

Strategy 03

Prioritise expansion in high-potential store clusters

Data Engineering (Source to Pipeline)

Source

- Dataset: Internal Chocolate Retail Simulation Dataset
- Files: calender.csv, customers.csv, products.csv, sales.csv, stores.csv
- Size: >5,000 sales transactions
- Time Period: Full calendar year

GitHub (notebooks)

/notebooks/01_extraction.ipynb
/notebooks/02_Cleaning.ipynb



Data Dictionary (key columns)

sales.csv: SaleID, Date, CustomerID, ProductID, StoreID, Boxes, Amount

customers.csv: Customer, Gender, Age

Python ETL (major cleaning steps)

1 Null handling

Dropped transactions with missing Amount/Boxes; validated all foreign keys

2 Type casting

Parsed dates to datetime64; standardised categorical fields

3 Deduplication

Verified unique SaleID; removed duplicate records

KPI & Metrics Framework

Total Revenue

25.2M

$\text{SUM}(\text{Amount})$

Total Orders

1M

$\text{COUNT}(\text{SaleID})$

Total Profit

10.1M

$\text{SUM}(\text{Profit})$

Profit Margin

40%

$\text{Total Profit} / \text{Revenue}$

Avg Order Value

25.5

$\text{Revenue} / \text{Orders}$

Revenue per Box

Calculated

$\text{SUM}(\text{Amount}) / \text{SUM}(\text{Boxes})$

Gross Margin %

Formulaic

$(\text{Price} - \text{Cost}) / \text{Price} \times 100$

Monthly Growth

MoM %

$\text{Revenue Change} \%$

? Why These KPIs

Directly answer revenue drivers and profitability across segments

</> Computation

notebooks/04_statistical_analysis.ipynb

notebooks/05_final_load_prep.ipynb

Key Insights (EDA) —



Revenue peaked in January

Seasonal spike indicates gifting season opportunity



Profit margin is 40%

Healthy profitability across portfolio



55+ age group is the top revenue driver

Older demographics outspend younger segments



Loyalty program needs improvement

Loyal and non-loyal customers spend almost equally



Canada is the top-performing country

Geographic concentration of revenue



Airport stores generate 29.87% of revenue

Highest revenue channel among store types

Advanced Analysis —

Analytical Depth

- Customer segmentation by Age Band (<18, 18–24, 25–34, 35–44, 45–54, 55+)
- Store performance ranking and benchmarking
- Repeat purchase rate analysis (IsRepeatCustomer flag)
- Gross margin analysis by product category

New Understanding



Dark Chocolate category drives disproportionate revenue despite fewer SKUs



Airport channel outperforms other store types



55+ demographic is underserved relative to spending power

Python Libraries Used:

pandas

numpy

matplotlib

seaborn

scipy

statsmodels

Tableau Dashboard Walkthrough



Executive View (Sales Dashboard)

KPI Scorecard

Orders

1M

Revenue

25.2M

Profit

10.1M

Sales Trend (Line)

Category (Pie)

Top Chocolates (Bar)

Top Stores (Bar)



Operational View

Customer Dashboard

50,000 Customers | Avg Revenue: 504.8 | Avg Age: 44.0 | Loyalty: 2.535%

Treemap

Line

Bar

Donut

Store Dashboard

100 Stores | 7 Cities | Top: Canada | Avg Store Profit: 10.20

Map

Donut

Treemap



Interactive Filters:

Category

Month

Store Type

Country

Age Group

Gender

Tableau Public URL

https://public.tableau.com/views/Chocolate_Sales_G17/SalesDashboard?:language=en-US&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

Recommendations

Insight	Recommendation	Expected Impact
1) 55+ age group drives highest revenue	Targeted marketing campaigns and premium product lines for 55+ demographic	+10-15% Revenue
2) Airport stores generate 29.87% of revenue	Prioritize inventory allocation and promotional spend for Airport locations	+5-8% Growth
3) Loyal and non-loyal customers spend equally	Redesign loyalty program with tiered benefits to incentivize repeat purchases	+15% Retention
4) January revenue peak (seasonal)	Build 6-week pre-Q1 inventory buffer; negotiate supplier capacity by November	~3% Recovered
5) Canada is top-performing country	Pilot premium gifting bundles exclusively in Canadian stores	+5% AOV

Impact, Limitations & Next Steps

Impact & Value



Estimated revenue uplift: 15–25% via targeted segmentation and channel optimization



Cost savings: Eliminate underperforming SKUs and optimize inventory



Efficiency: Data-driven promotional budget allocation

Why Stakeholder Should Act:

- Cost of inaction: Continued suboptimal marketing spend; missed revenue from underserved demographics
- Competitive advantage: First-mover in targeted 55+ chocolate market

Limitations

01

Currency not specified in dataset (all monetary figures treated as consistent unit)

02

Some CustomerID values unmatched – ~X% excluded from customer-segment analysis

03

Negative Amount values (returns) excluded from revenue KPIs

04

Dataset is simulated retail data, not real-world transactions

Next Steps

Predictive Modeling

Build predictive model for customer lifetime value

Real-time BI

Develop real-time dashboard for inventory management

Sentiment Analysis

Extend dataset with actual customer feedback and sentiment data

A/B Testing

A/B test loyalty program redesign

Team & Contribution Matrix



Team Member	Dataset & Sourcing	ETL & Cleaning	EDA & Analysis	Statistical Analysis	Tableau Dashboard	Report Writing	PPT & Viva
Yashkumar Nimje	✓				✓		
Shane Joseph		✓				✓	
Manpreet Singh				✓	✓		
Shivansh Upadhyay			✓	✓			
Saksham Narotra							✓
Ashar Shaikh							✓



GitHub contribution history as evidence

Thank You

Q&A